

Welcome to Deep Learning for Business IS 695 Sec 01

WE 7:00PM – 9:45PM

DA XU

Course Description

This course introduces

- Fundamental information theories and methodologies in data analytics and deep learning
- Core deep learning techniques and algorithms for preprocessing and analyzing a variety type of large-scale data sets
- Hands-on practices of major deep learning techniques with multiple real-world business applications enabled by deep learning.
- Both fundamental theories and hands-on experiences are emphasized in this course.

Materials

- Textbook (Optional): Stevens, Eli, Luca Antiga, and Thomas Viehmann. Deep learning with PyTorch. Manning Publications.
- Course materials are available on BeachBoard
- Software
 - Google Colab for Python programming

Technical Assistance

- If you need technical assistance at any time during the course or need to report a problem with BeachBoard, please contact the Technology Help Desk using their online form - [AT Help Form Homepage](#) or by phone at (562) 985-4959 or visit them on campus in the Horn Center or in the Library.

About This Course

- This is basically an introduction to deep learning where you will be exposed to a number of different concepts, methods, and algorithms, and learn to apply them to solve business problems.

Grading

Grading*

	Points	Percentage
2 Exams (75 points each)	150	30%
2 Quizzes (25 points each)	50	10%
10 Lab Assignments (10 points each, drop the lowest)	100	20%
4 Homework Assignments (25 points each)	100	20%
1 Group Project	100	20%
Total	500	100%

*Notes:

1. The number of assignments for each category may change.
2. Detailed description of each assignment will be available on BeachBoard.
3. All Quizzes and Exams are open book.

For each lab, you will get 10 points if you submit a complete lab file by the due date.

Grading Scale

Grading Scale

Percent Range	Letter Grade
90-100%	A
80-89%	B
70-79%	C
60-69%	D
Below 60%	F

BeachBoard

Dates and Schedules

- Pay attention the class meeting dates and due dates

News

- BeachBoard News will be made from time to time, please pay attention to them

The “week” module

- It is very important to check the "week" module and be clear about the weekly tasks and due days

Communication

Primary communication method will be email

- da.xu@csulb.edu

Office Hours

- Tu/Th 3:15PM - 4:15PM

- Tu/Th 6:00PM - 7:00PM

- **Office:** COB-439

Attendance

- No portion of your grade will be tied directly to attendance, however, class meetings and participation are vital components of this course
- A large portion of your grade is based on in classroom activities
 - Quizzes, and Exams will be completed on class
 - All material covered in class is expected to be used on the exams
 - Labs

Assignments

- All homework and lab assignments are due at 11:59 pm on the due date as posted in BeachBoard
- For late assignments (up to 3 days late), the final assignment score will be dropped by 30% unless prior approval for an extension has been obtained
- **Do not wait** until the last moment to submit on BeachBoard

Quizzes and Exams

- All Quizzes and Exams are open book
- Quizzes and Exams include
 - Part1: True/False and multiple choice questions
 - Part2: Short answers
 - Part3: Code writing – you will be required to write and submit Python codes on BeachBoard.

Question?

Lecture 1: Introduction to Deep Learning

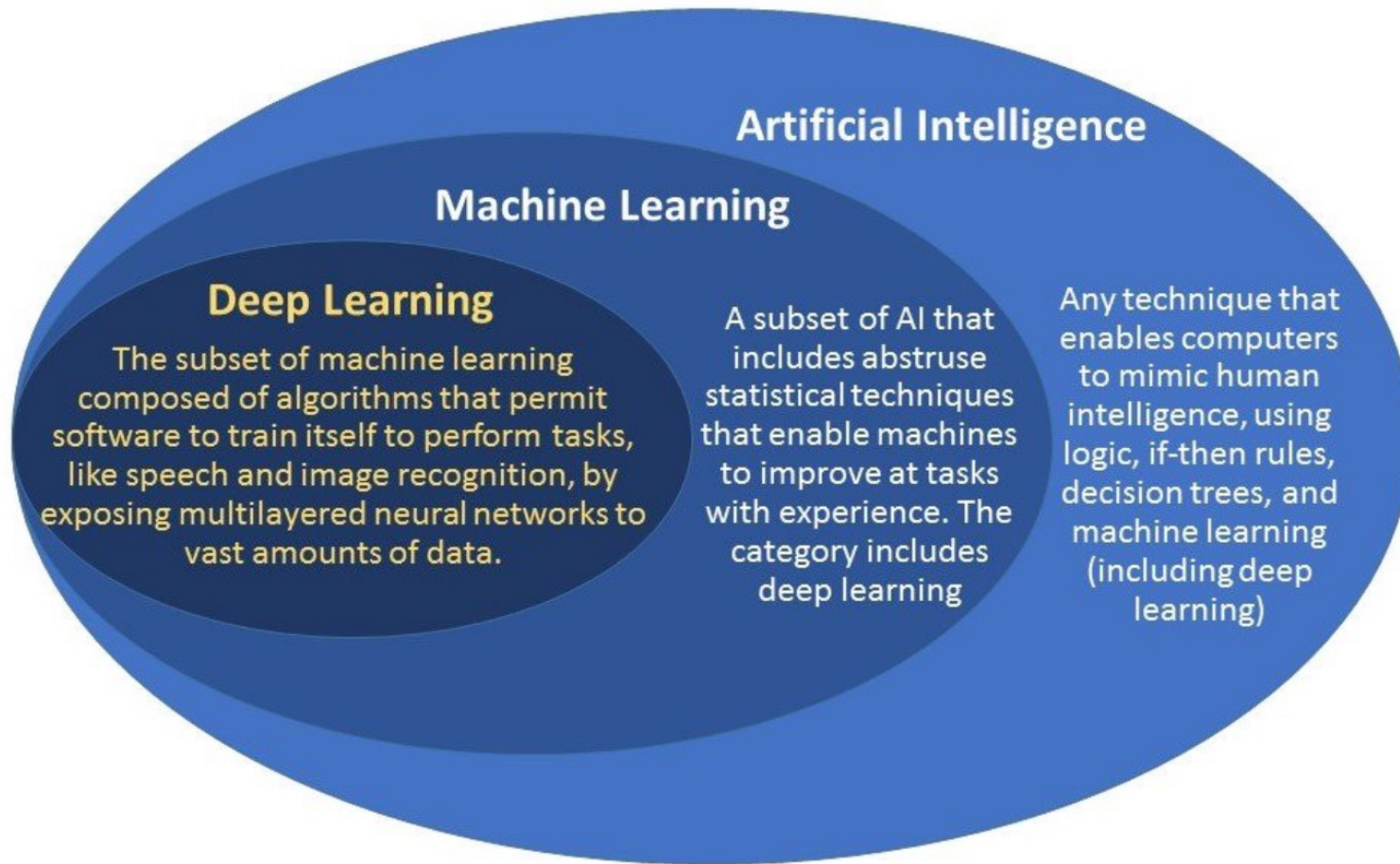
Overview

- Machine Learning
- What is Deep Learning
- Why Deep Learning
- Deep Learning Advantages
- Deep Learning Applications
- Deep Learning Trends

Machine Learning

- Machine learning is a branch of artificial intelligence based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.
- Deep learning (also known as deep structured learning, hierarchical learning or deep machine learning) is a class of machine learning algorithms.
 - Deep learning algorithms: neural networks

Machine Learning



Why Machine Learning?

- Data: data are any facts, numbers, or text that can be processed by a computer.
 - Computerization of businesses produce huge amount of data
 - Online e-businesses are generating even larger data sets

Why Machine Learning?

- Data



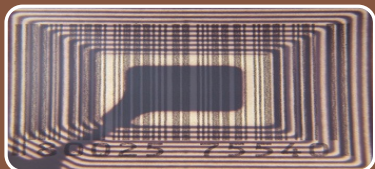
Business data – e.g., demographic data, transactional databases, and customer behavior data



Web data – e.g., page content, page linkage and clickstream



System data – e.g., service logs, quality and availability of network, computer, and manufacturing devices



Sensor data – e.g. RFID devices for homeland security, supply-chain and environmental applications

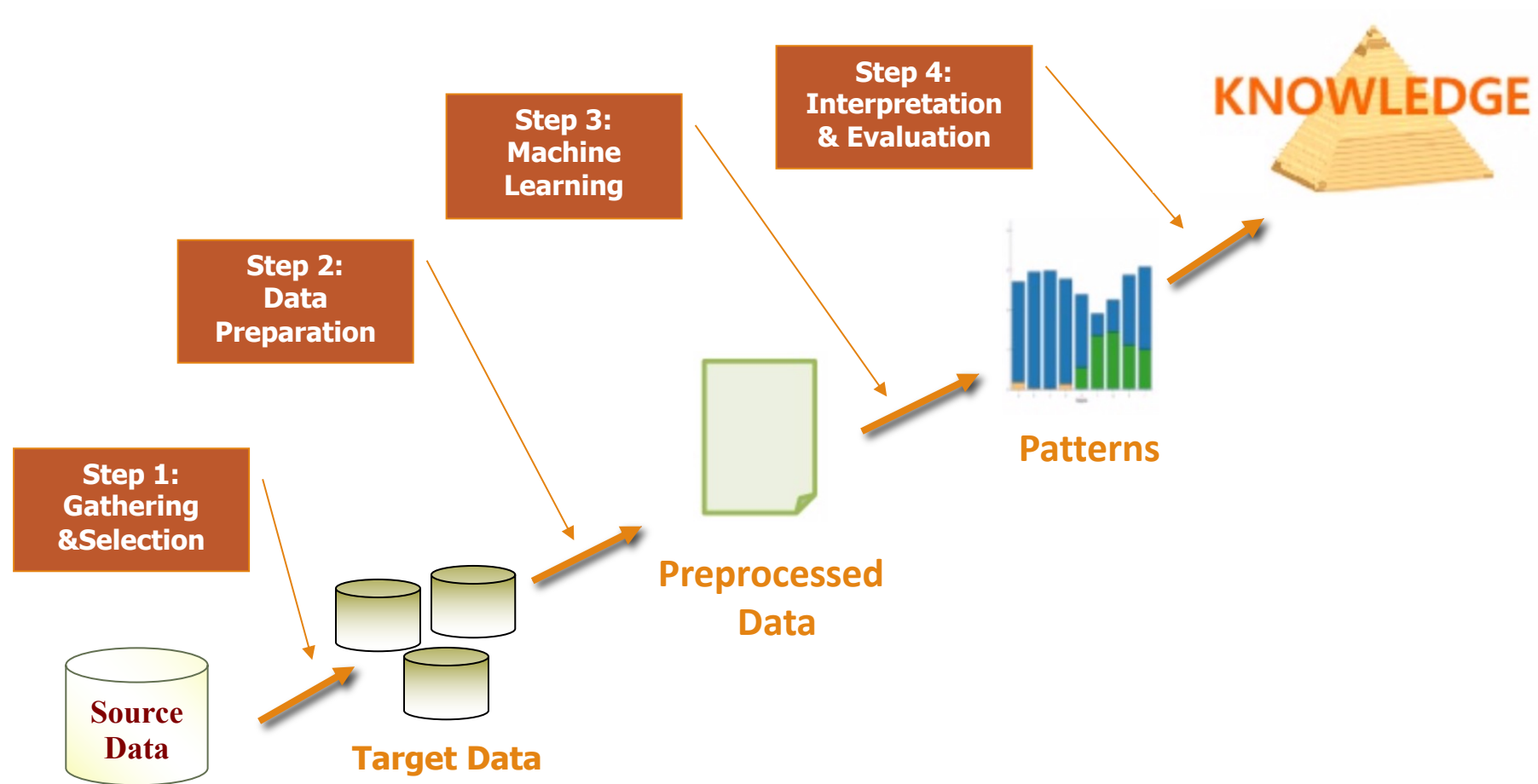
Why Machine Learning?

- There is a huge gap from stored data to knowledge

Why Machine Learning?

- Machine Learning helps generate insights from data to support business decisions and improve the profitability and efficiency

Machine Learning Process



Machine Learning Process

- **Data understanding:** With preliminary analysis, data exploration provides a high level overview of each variable in the data set and interaction between the variables.

The diagram illustrates a data table. A bracket above the columns is labeled 'features', and a bracket to the right of the rows is labeled 'examples'.

year	model	price	mileage	color	transmission
2011	SEL	21992	7413	Yellow	AUTO
2011	SEL	20995	10926	Gray	AUTO
2011	SEL	19995	7351	Silver	AUTO
2011	SEL	17809	11613	Gray	AUTO
2012	SE	17500	8367	White	MANUAL
2010	SEL	17495	25125	Silver	AUTO
2011	SEL	17000	27393	Blue	AUTO
2010	SEL	16995	21026	Silver	AUTO
2011	SES	16995	32655	Silver	AUTO

If a variable represents a characteristic measured in numbers, it is unsurprisingly called **numeric variable**.

if a variable consists of a set of categories, the variable is called **categorical variable**.

Machine Learning Process

- **Data preparation:** Before applying the machine learning algorithm, we need to prepare the data set for handling of any anomalies that may be present in the data.
 - Finding outliers; e.g., patients with age >120
 - Missing values; data sparsity
 - Duplicate
 - Highly correlated variables; age/DOB
 - Data cleaning and transforming

Machine Learning Process

- **Machine learning tasks**
 - Classification
 - Numeric prediction
 - Clustering
 - Association rule mining

Machine Learning Process

- **Interpretation and evaluation:**

- Because each machine learning model results in a biased solution to the learning problem, it is important to evaluate how well the algorithm learns from its experience.
- Depending on the type of model used, you might be able to evaluate the accuracy of the model using a test dataset or you may need to develop measures of performance specific to the intended application.

From Biological to Artificial Neurons

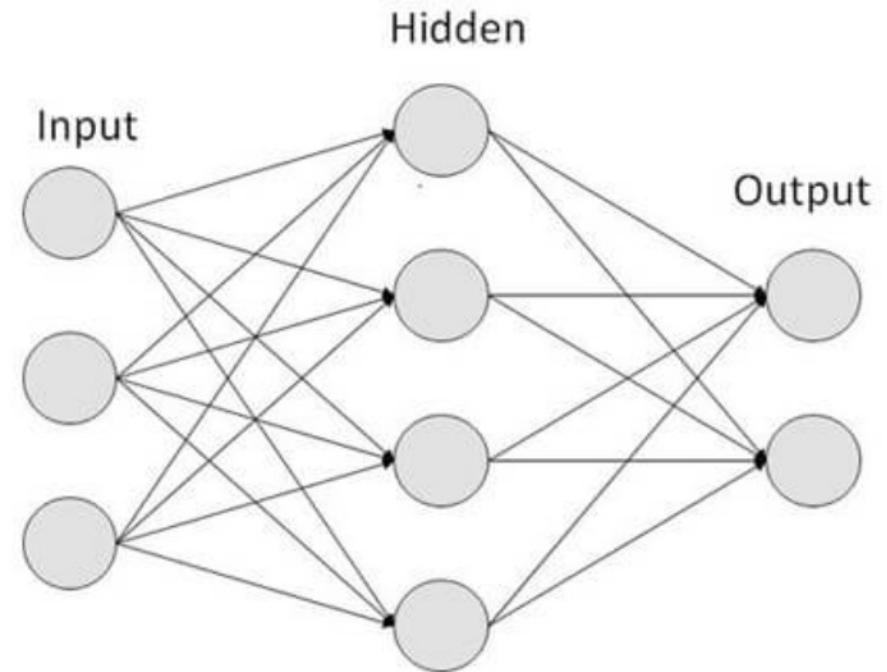
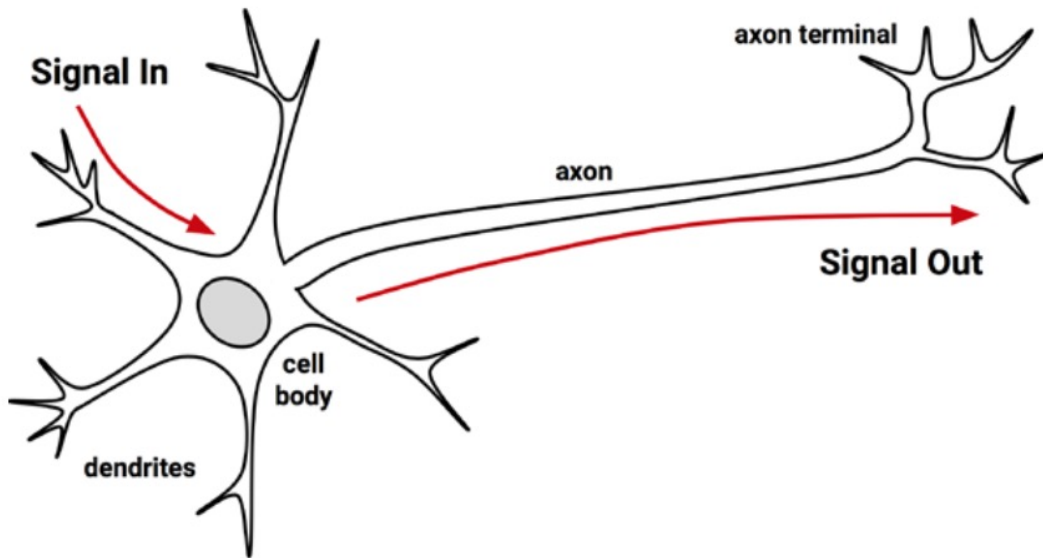
- Neural networks, are computational models inspired by biological neural networks, particularly, they are inspired by the behavior of neurons and the electrical signals they convey between input, processing, and output from the brain

From Biological to Artificial Neurons

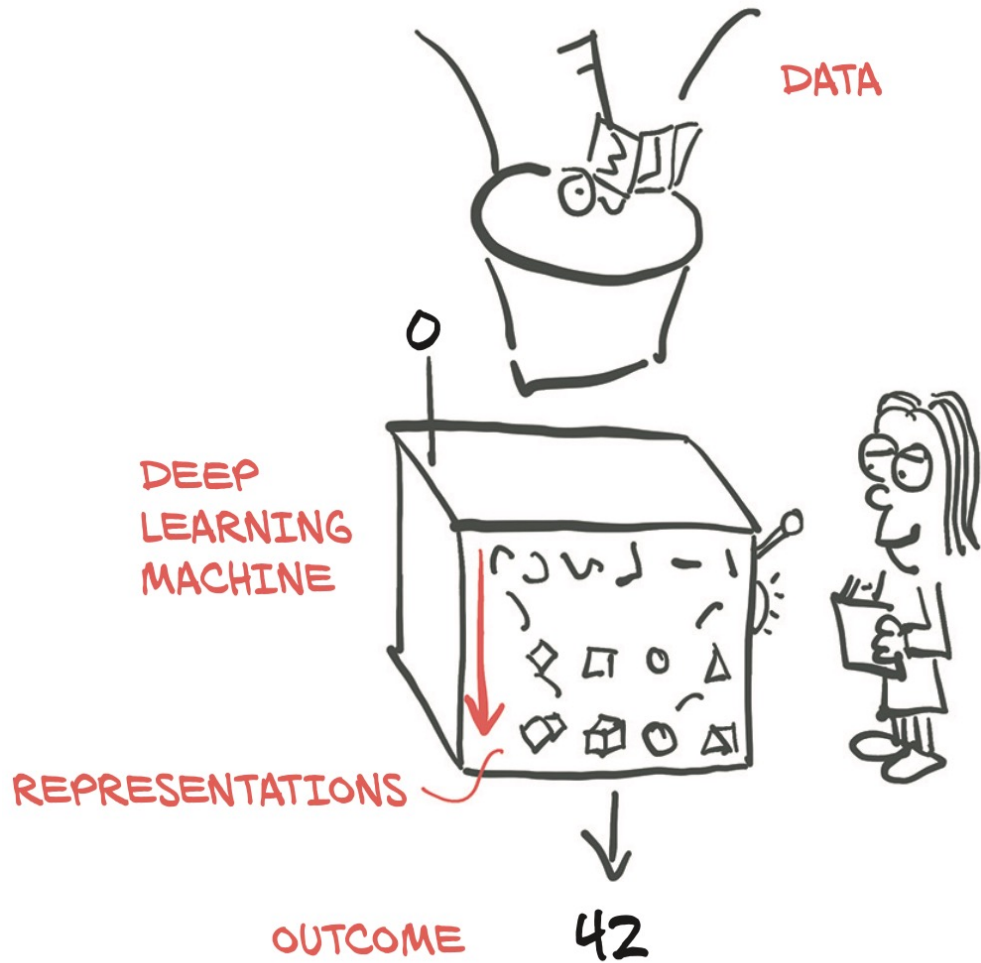
- An Artificial Neural Network (ANN) models the relationship between a set of input signals and an output signal
 - Derived from our understanding of how a biological brain responds to stimuli from sensory inputs
 - A brain uses a network of interconnected cells called **neurons** to create a massive parallel processor
 - ANN uses a network of **artificial neurons** or **nodes** to solve learning problems
- ANNs are versatile learners that can be applied to nearly any learning tasks
 - Classification, numeric prediction, and even unsupervised pattern recognition

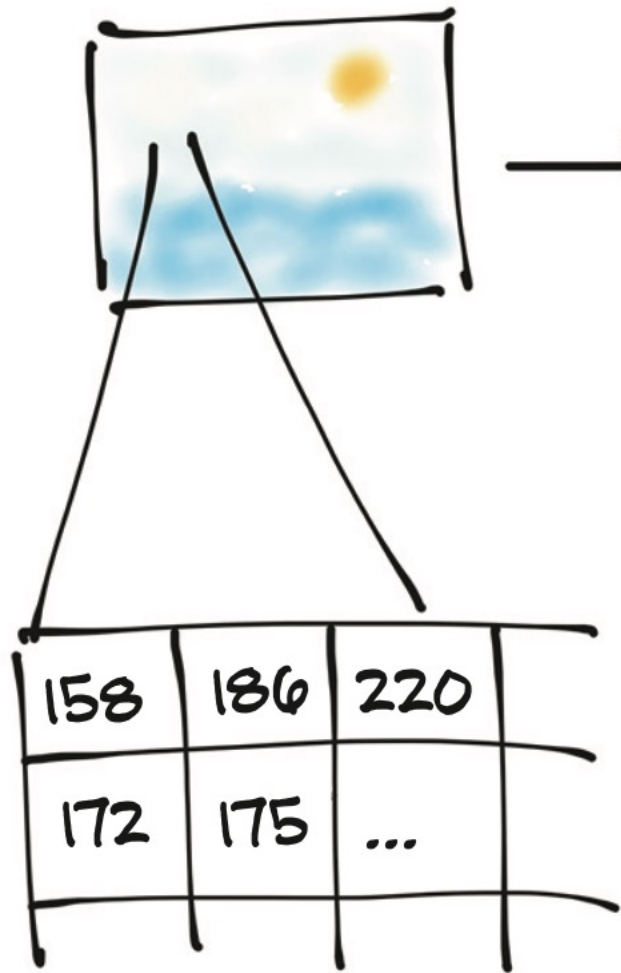
From Biological to Artificial Neurons

- Nature has inspired millions of innovations-the human brain has inspired the creation of neural networks.

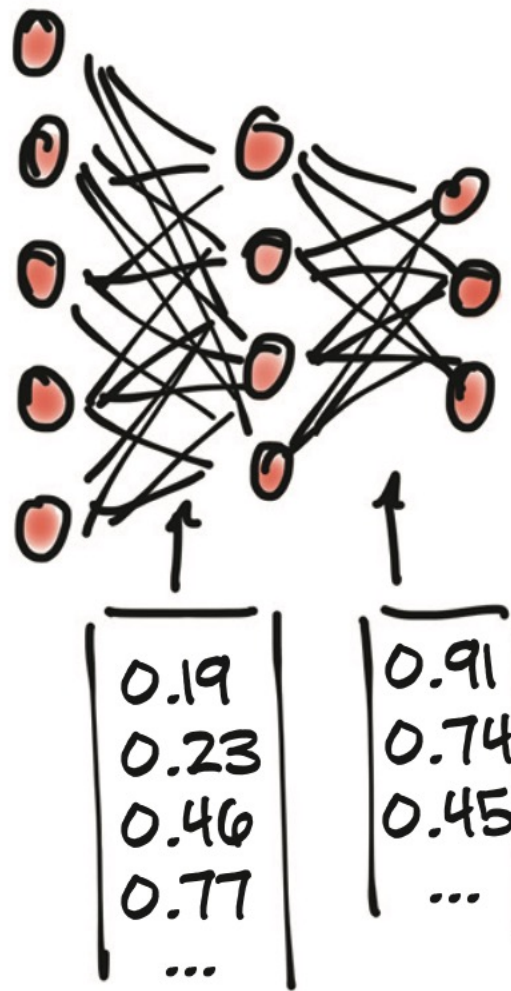


Deep Learning

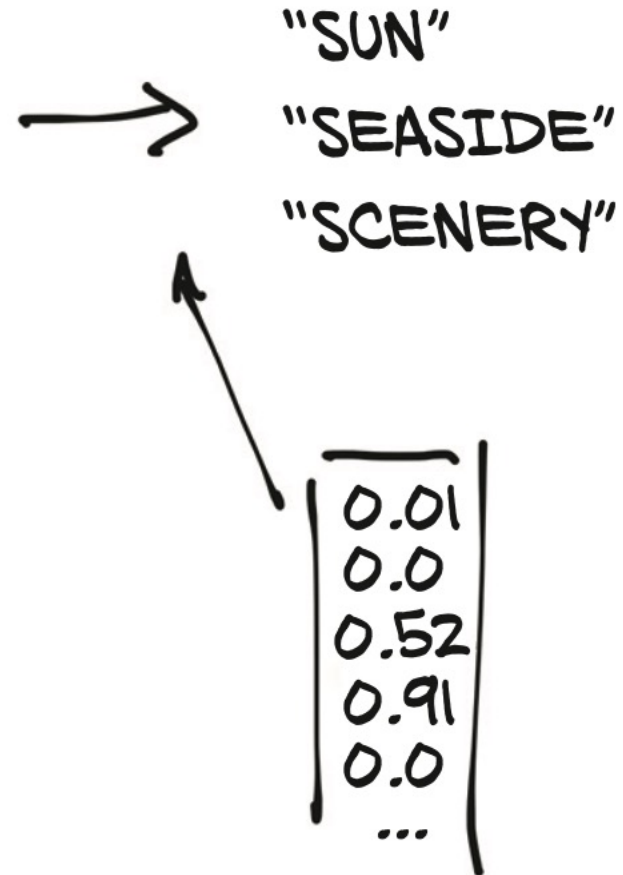




INPUT
REPRESENTATION
(VALUES OF PIXELS)

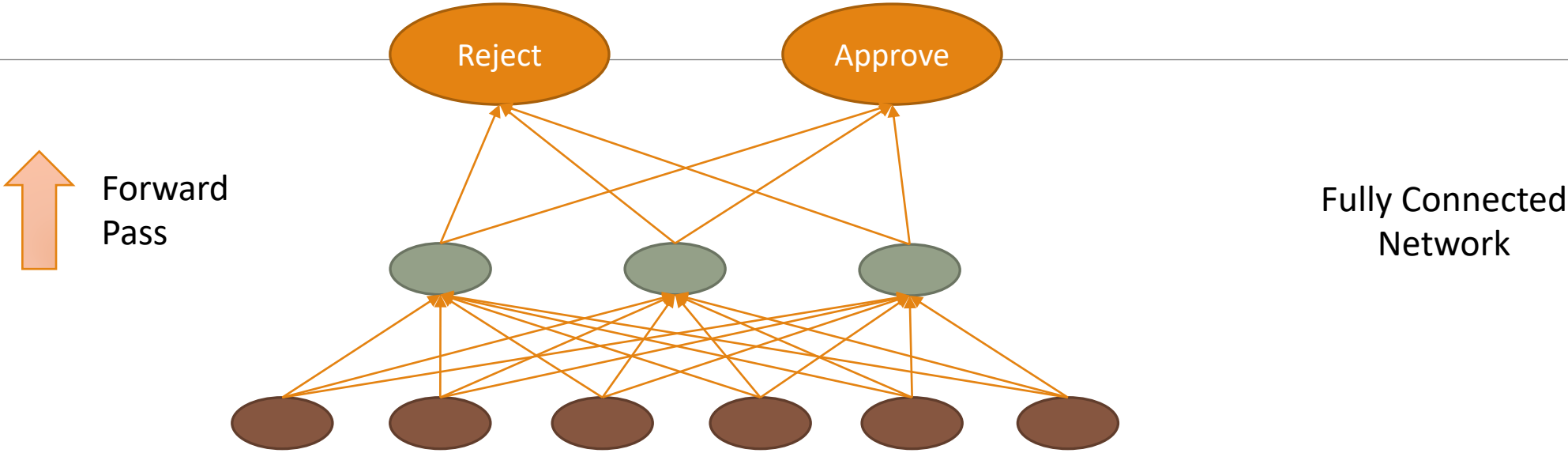


INTERMEDIATE
REPRESENTATIONS



OUTPUT
REPRESENTATION
(PROBABILITY OF CLASSES)

Neural Network



CID	Credit Score	Income (10,000)	Term	Age	Gender	Number of Children	Status
1	6	2	1	35	0	0	Reject
2	7.5	5	3	38	0	2	Approve
3	4	3.5	2	23	1	4	Reject
...							
100000	8	2.5	1	35	0	1	Approve

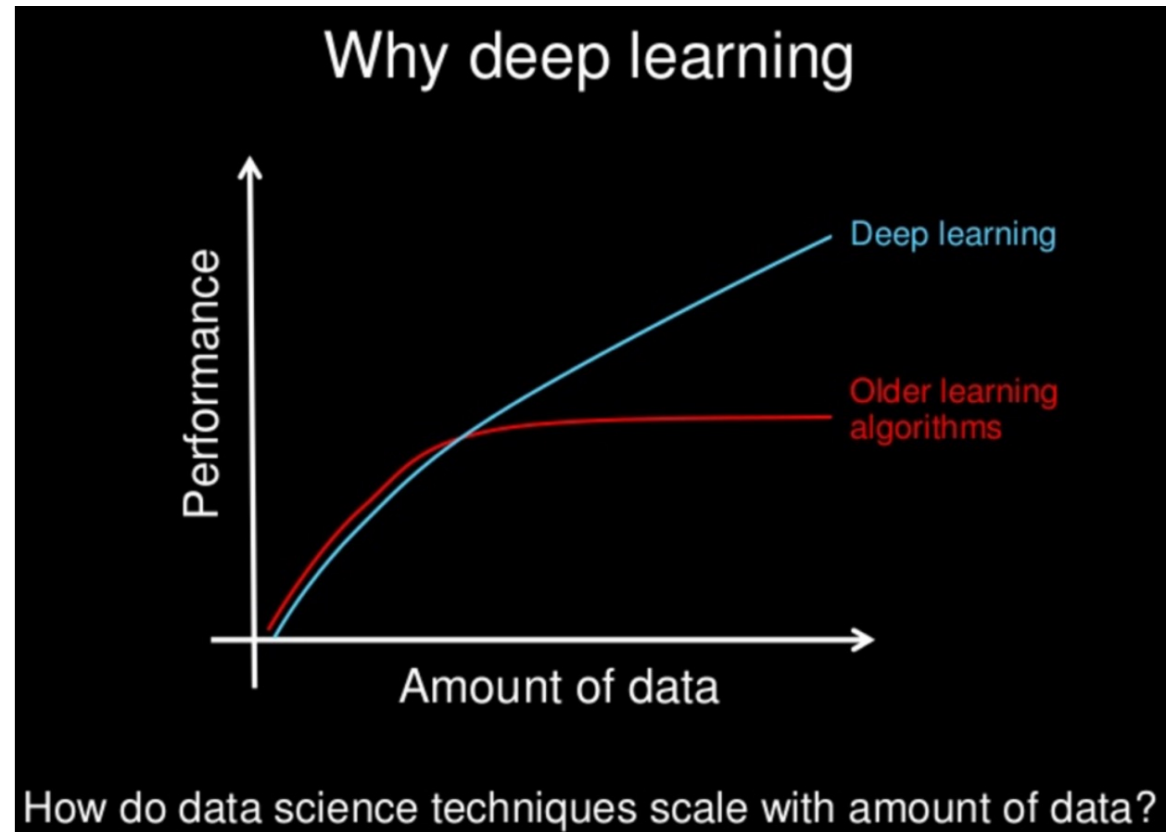
Why Deep Learning

■ Advantages:

- State of art performance
- Capability of handling unstructured data
- Automatic feature engineering

■ Disadvantages:

- More data
- Bigger models; hard for model training and tuning
- Difficult to interpret
- Resources demanding



Why Deep Learning

- Different types of data

Class Roster			
Course:	MGT 500 Business Policy	Semester:	Spring 2010
Section:	2		
<u>Name</u>	<u>ID</u>	<u>Major</u>	<u>GPA</u>
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Structured data: data stored in tabular form



Unstructured data: image, video, audio, text

Techniques of Deep Learning

- Different types of data

- Feedforward neural network: deep learning model with fully connected layers.
- Convolutional neural network (CNN): a specialized kind of neural network for processing data that has a known, grid-like topology (e.g., image data).
- Recurrent neural network (RNN): a family of neural networks for processing sequential data (e.g., text data).

Application – Recommender Systems

- Data knows more about your music preference



Application – Image Processing

■ Image classification

Photos for The Cheesecake Factory



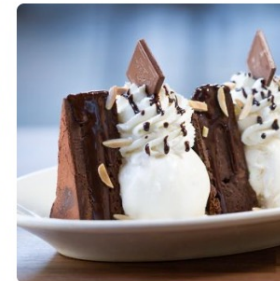
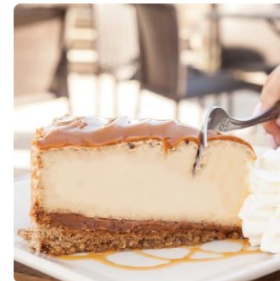
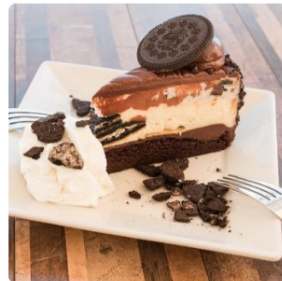
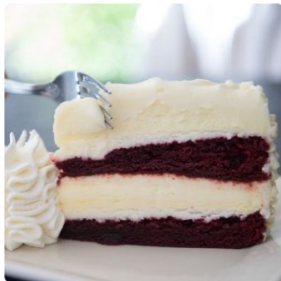
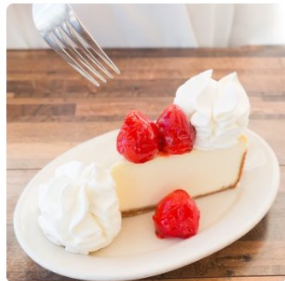
The Cheesecake Factory

★★★★☆ 472 reviews

 Add photos

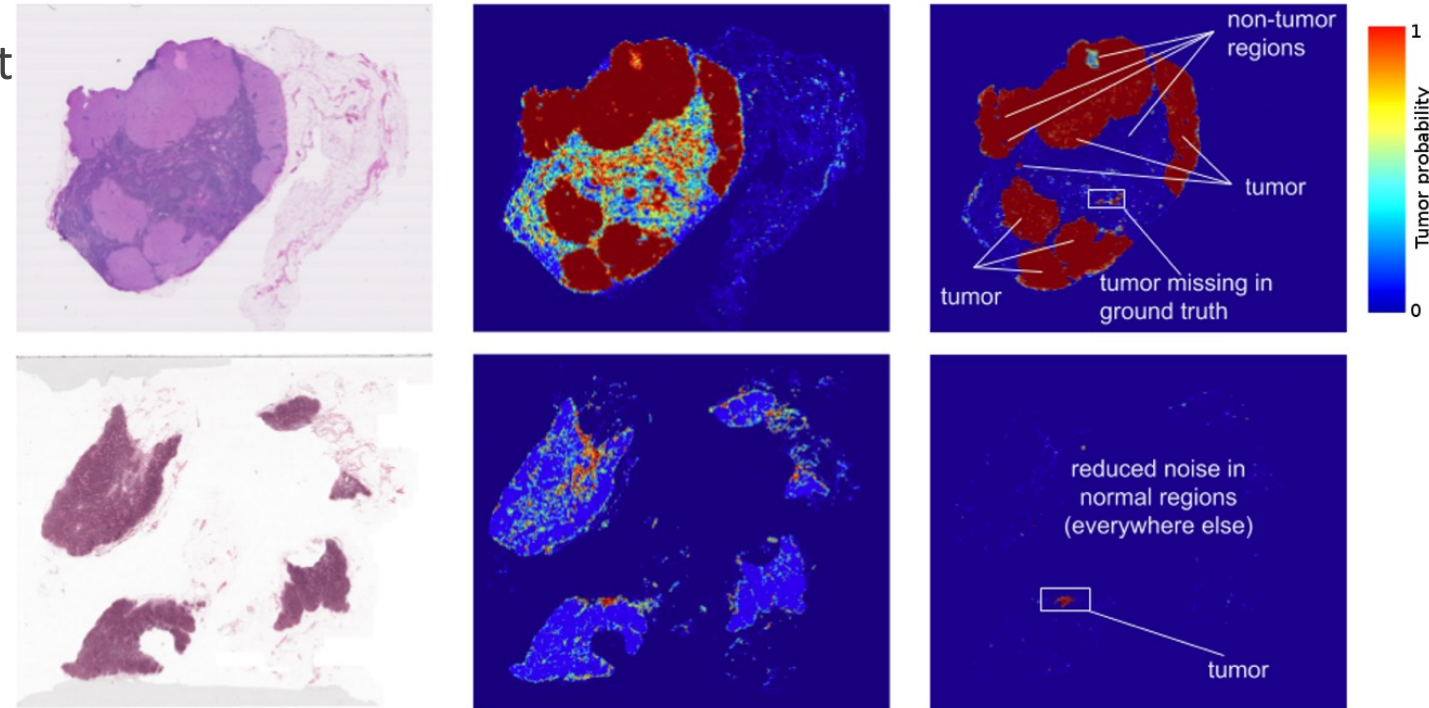
< **All (537)** Food (453) Inside (29) Drink (20) Menu (8) Outside (7)

> Search photos...



Application – Image Processing

- Researchers at IBM estimate that medical images currently account for at least 90 percent of all medical data, making it the largest data source in the healthcare industry.
- **Inception v3** by Google is an example of AI-based image analysis that works with DNA and cancer data.
 - Developers used thousands of images of both healthy and cancerous tissues. **Inception v3** was able to advance on its own and learn the difference between the two types of tissues.
 - Inception v3 can also find mutations inside the DNA of cancer cells, helping doctors discover mutations and get ahead of the disease with appropriate treatment



The picture shows that deep learning-based image analysis can detect lymph node tumor

Application – Image Processing

- Tesla grants certain customers early access to a “feature complete” version of the company’s “full self-driving” (FSD) capabilities in 2020.
 - The car is able to drive from someone’s home to their work without intervention.

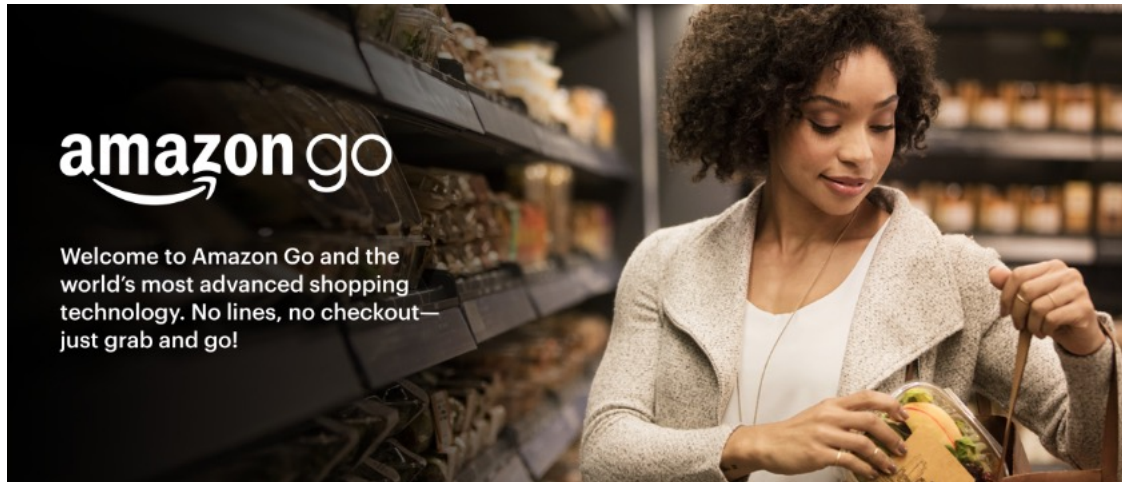
How self-driving cars work?

- Self-driving cars combine a variety of sensors to perceive their surroundings.
- Advanced control systems interpret sensory information to identify appropriate navigation paths, as well as obstacles and relevant signage.



Application – Image Processing

Amazon Go is an automated grocery stores that customers are able to purchase products without being checked out by a cashier or using a self-checkout station.



Go technology, a complicated array of cameras and software, figures out what shoppers have grabbed and automatically charges them when they exit.

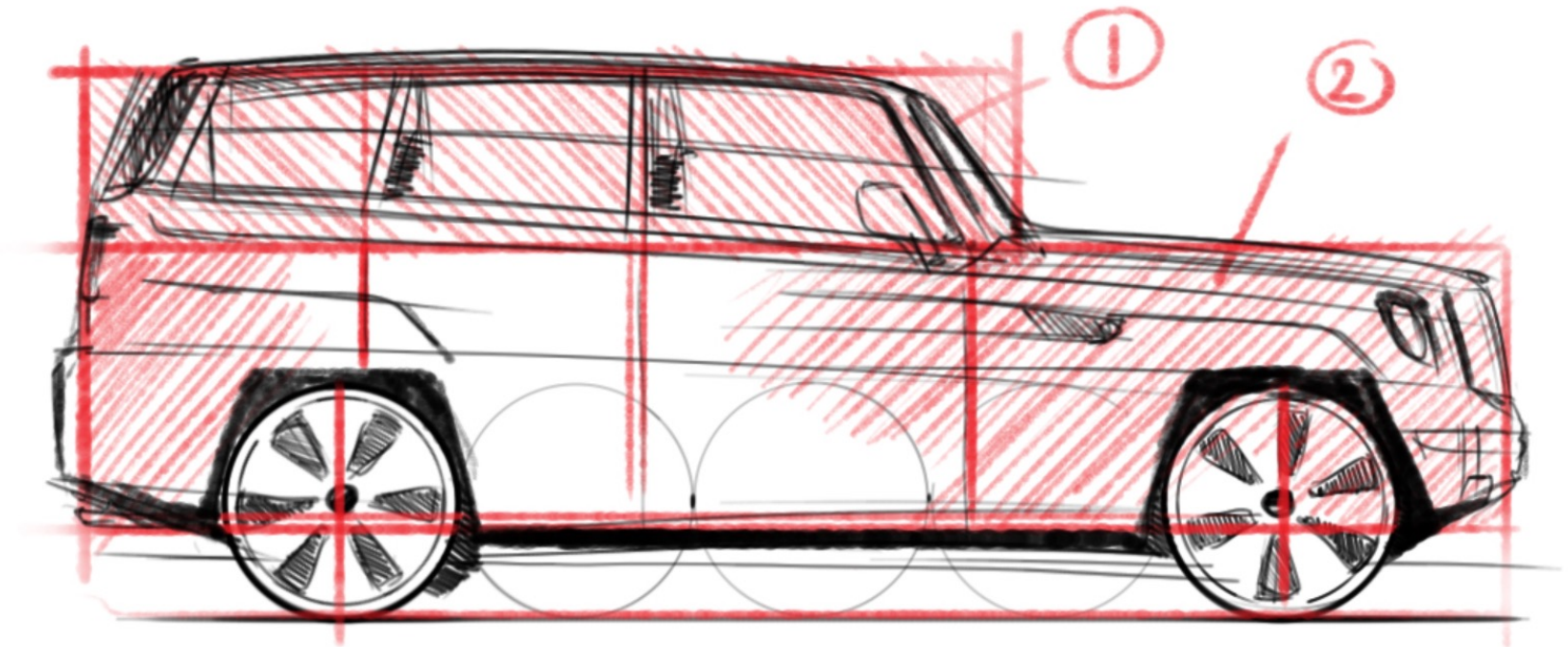
Application – Image Processing

- Amazon is reportedly eyeing to open 3,000 Amazon Go stores by 2021 and is said to be looking at expanding the concept to London, airports and colleges.
- The Amazon Go team will soon expand and apply the Go technology to Wholefoods and other supermarkets.
- The company will open much larger locations equipped with its technology, which range from 1,500 to 3,000 square feet.



Application – Natural Language Processing

- Car manufacturer improved vehicle models via the social media platform



Application – Natural Language Processing

- Google Translate mobile app has the instant camera translation, which allows you to see the world in your language by just pointing your camera lens at the foreign text.

How Google translate works?

- Google's AI translation methods is known as neural machine translation (NMT).
- These models have been incorporated into Google Lens and the web version of Translate, but they are now supporting instant camera translation.



Application – Natural Language Processing

- Early in 1998, Wuthrich et al. attempted to predict stock markets based on online news articles such as The Wall Street Journal.
- Now, the natural language processing community correlates the text contents with the sentiment (positive or negative) of financial forums posts to predict the stock prices.



Another type of study is about applying social media text mining for stock market prediction.

- The method leverages market information contained in high-volume social media data rather than news articles.

Application – Generative Model

- Today, AI can create realistic images and videos of cats, products, faces that aren't of real people, your voices and even original works of art.



2014



2015



2016



2017



2018

- Each of the faces in the image from the left was generated by an AI.
- Starting with the faces on the left, from 2014, you can see how dramatically AI capabilities have improved.

Application – Generative Model

- Deep Learning
 - Generative adversarial network (GAN) — a class of neural network — to generate some extremely realistic faces.



Application – Generative Model

- Deep Learning
 - Generative adversarial network (GAN) — a class of neural network — to generate some extremely realistic faces.



Application – Generative Model

- In fact, there's even a market for AI's original artwork—Google hosted an art show to benefit charity and to showcase work created by its software DeepDream. It sold an AI-generated piece of art that was a collaboration between human and machine for \$8,000 plus others.

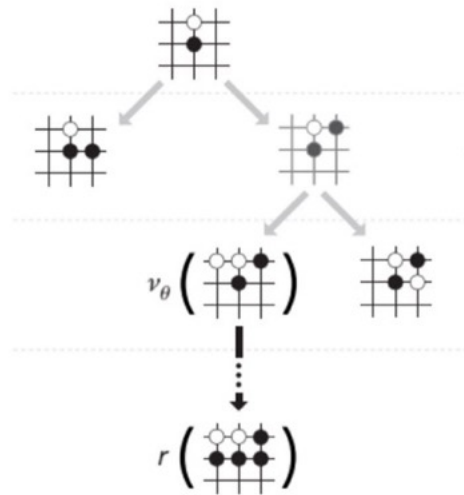


British auction house Christie sold its first piece of AI-generated art, “Portrait of Edmond Belamy,” for \$432,500.



Application – Reinforcement Learning

- Reinforcement Learning is quite widely used in building AI for playing computer games.
- **AlphaGo** and **AlphaGo Zero** are the first computer programs to defeat a world champion in the ancient Chinese game of Go.
- In the Go game, the algorithm searches the possible moves and finds the actions to win the game. Of course, the search space is too large and need to search smarter.



Application – Reinforcement Learning

- AlphaStar: Mastering the Real-Time Strategy Game StarCraft II
- During the training, each agent experienced up to 200 years of real-time StarCraft play.

